

Investigating the Wedge Map on SNOVA

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Abstract. This paper investigates the algebraic structure of SNOVA, a Round 2 candidate of NIST PQC competition for additional signatures, with a specific focus on the kernel dimension of the wedge map. We employ lifting techniques to transform the public key from the matrix ring $\mathbb{F}_q^{l \times l}$ to an equivalent representation over the extension field $\mathbb{F}_{q^l}$, establishing that the rank of the wedge map is a structural invariant. A key contribution of this work is the derivation of a generating function that explicitly characterizes the wedge map’s kernel dimension. This algebraic analysis provides a rigorous understanding of SNOVA’s geometry, verifying the safety of SNOVA parameter sets and enabling future refined adjustments to guarantee structural security.

1 Introduction

In response to the critical threat that quantum computers pose to public-key cryptographic systems, the U.S. National Institute of Standards and Technology (NIST) has launched a series of efforts toward the standardization of post-quantum cryptography (PQC). In recent years, significant research attention has been devoted to the NIST Standardization of Additional Digital Signature Schemes competition (onramp process). Among the Round 2 candidates, schemes belonging to the UOV family have attracted considerable interest. Owing to its simple structure and extensive prior study, UOV [3] is widely regarded as one of the most promising candidates. Meanwhile, other UOV variants: SNOVA [17,18], MAYO [2] and QR-UOV [6] employ different mathematical techniques to overcome UOVs main drawback, namely its excessively large public key size. SNOVA [17] is a Round 2 candidate in NIST ongoing onramp process for additional digital signature schemes. SNOVA builds on the UOV [3] scheme, modifying its structure to significantly reduce public key size and improve signing efficiency, while preserving the advantages of short signatures and fast verification. For example, at NIST security level I, SNOVA achieves public keys as small as 1016 bytes and signatures of 248 bytes, while maintaining efficient signing and verification performance. Due to its compactness and efficiency, SNOVA represents

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an attractive option for practical real-world deployments. Therefore, numerous security analyses [10,8,1,12,15,16,4,14,7] of SNOVA have been presented in the literature.

In [13], Lars Ran pointed out that the polar form of the UOV public map is not only symmetric but also alternating in characteristic 2. By viewing these polar forms as elements of the exterior algebra, this observation leads to a new key-recovery attack on UOV [3] and its variants. Ran further noted that the underlying system within SNOVA [17] exhibits additional structural properties, suggesting that the complexity estimates provided in his work may not be fully accurate for SNOVA. Following this, Le *et al.* [9] presented an estimation of the wedge attack specifically targeting SNOVA. They analyzed the wedge attack against SNOVA in their paper. In particular, they proposed two approaches to estimate the rank of the wedge map (or equivalently, the dimension of its kernel). Through experimental results on small parameter sets, they showed that the actual rank typically lies between the values predicted by these two estimation approaches.

From the perspective of the wedge attack on SNOVA, both [13,9] leave two important questions that still require clarification:

- How should the wedge attack on SNOVA be properly formulated and explained?
- How to correctly predict the dimension of the kernel of the wedge map?

Once we can fully and precisely resolve the two questions mentioned above, we will be able to provide a concise and efficient framework for describing the wedge attack on SNOVA. Moreover, by accurately estimating the dimension of the kernel of the wedge map, we can derive the correct complexity of the wedge attack for given SNOVA parameter sets.

The value of such security analysis lies in the following aspects: It provides new progress on wedge attacks within the UOV family that completes the study of the wedge attack against SNOVA. It enables more precise parameter adjustments of SNOVA scheme.

Our contributions. The main contributions of this work are as follows:

- We provide a proper formulation of the wedge attack on SNOVA and reveal the underlying structure of the wedge map in Section 3.
- We derived a generating function that accurately predicts the kernel dimension of the wedge map in the case of SNOVA, correcting previous heuristic formulas in the literature [9] in Section 4. In Section 4, we also provide the complexity estimates of SNOVA parameters under the wedge attack together with some potential parameter adjustments.
- We provide a mathematical proof and verification of the estimation of kernel dimension of the wedge map, based on the computation of homological algebra in Section 5.

We provide a comprehensive algebraic analysis of SNOVA which is based on the natural isomorphic relation between the scheme's original matrix ring structure and its representation over an extension field. By employing explicit basis

changes, we transform the public key into a form that highlights the underlying complex, thereby proving that the wedge map's rank is a preserved structural invariant. A central achievement of this work is the derivation of a generating function that yields an accurate prediction of the wedge map's kernel dimension, effectively correcting previous heuristic formulas that fail to correctly predict the actual rank [9].

The precise prediction presented in this paper offers a distinct advantage in security evaluation. In this work, we believe that we have answered the two central questions of interest: establishing an appropriate formulation of the wedge attack on SNOVA, and providing an accurate prediction of the kernel dimension, together with strict mathematical justification. We believe that our result brings a positive impact on the adjustment of SNOVA parameters and ensures that the scheme's resistance to structural attacks is based on accurate estimates rather than rough approximations.

1.1 Notation

- (v, o, q, l, m) SNOVA denotes a derived SNOVA scheme under the wedge attack with v vinegar (ring) variables, o oil (ring) variables and m quadratic (ring) equations over $\mathbb{F}_q$. Let $n = v + o$
- $[c]$ denotes the set $\{1, \dots, c\}$.
- $\mathbb{F}_q$ denotes a finite field with q elements.
- $\mathbb{F}_q^c$ denotes the space of c -dimensional column vectors over $\mathbb{F}_q$.
- $\mathbb{F}_q^{r \times c}$ denotes the set of matrices of size $r \times c$ with entries in $\mathbb{F}_q$.
- $S \in \mathbb{F}_q^{l \times l}$ is symmetric matrix with an irreducible characteristic polynomial.
- $\mathbb{F}_q[S]$ denotes the set $\{a_0 S^0 + a_1 S + \dots + a_{l-1} S^{l-1} : a_0, \dots, a_{l-1} \in \mathbb{F}_q\}$.
- $\otimes$ denotes the Kronecker tensor product.
- $\tau : \mathbb{F}_{q^l} \rightarrow \mathbb{F}_{q^l}$ denotes the Frobenius map defined by $z \mapsto z^q$.
- $\mathbf{Q}^{\otimes n}$ denotes the block-diagonal matrix with n copies of Q along the diagonal.

2 SNOVA and Lifted SNOVA

SNOVA [18,17] is a multivariate signature scheme proposed by Wang *et al.* in 2023. It can be viewed as a UOV [3] variant defined over the l -by- l matrix ring $\mathbb{F}_q^{l \times l}$. This section describes the construction of Round 2 SNOVA [17] in the NIST onramp process and its lifted version.

2.1 SNOVA

In this subsection, we focus on the structure of the public and secret keys in SNOVA scheme. For a complete description of the signature generation and verification procedures, we refer the reader to the SNOVA Round 2 specification [17].

Let v, o denote the number of vinegar and oil variables, respectively. Let $n = v + o$ denote the number of variables and m denote the number of equations. Let q denote the size of finite field $\mathbb{F}_q$. Let $S \in \mathbb{F}_q^{l \times l}$ be a symmetric matrix with irreducible characteristic polynomial. The matrix S is given as a public parameter of SNOVA. We also define the subring $\mathbb{F}_q[S]$ of the $l \times l$ matrix ring $\mathbb{F}_q^{l \times l}$ by the following

$$\mathbb{F}_q[S] = \{a_0 + a_1 S + \cdots + a_{l-1} S^{l-1} : a_0, a_1, \dots, a_{l-1} \in \mathbb{F}_q\}. \quad (2.1)$$

The public key and private key of a (v, o, q, l, m) SNOVA scheme is constructed as the following:

Private key. The private key is given by an invertible matrix $\mathbf{T} \in (\mathbb{F}_q[S])^{n \times n}$ and the matrices $\mathbf{F}_1, \dots, \mathbf{F}_m \in \mathbb{F}_q^{ln \times ln}$. More precisely, we have

$$\mathbf{T} = \begin{bmatrix} I_{lv \times lv} & T^{12} \\ 0 & I_{lo \times lo} \end{bmatrix} \quad (2.2)$$

where $T^{12} \in (\mathbb{F}_q[S])^{v \times o}$ and for each $k \in \{1, \dots, m\}$,

$$\mathbf{F}_k = \begin{bmatrix} *_{lv \times lv} & *_{lv \times lo} \\ *_{lo \times lv} & 0 \end{bmatrix}. \quad (2.3)$$

Public Key. The public key of SNOVA is consisting of $\mathbf{P}_1, \dots, \mathbf{P}_m \in \mathbb{F}_q^{ln \times ln}$ where $\mathbf{P}_k = \mathbf{T}^\top \mathbf{F}_k \mathbf{T}$.

2.2 Lifting SNOVA

In this subsection, we review the technique of lifting the SNOVA scheme to the extension field $\mathbb{F}_{q^l}$ as introduced in [4,7] and clarify the relationship between SNOVA and its lifted version.

Let $\tau : \mathbb{F}_{q^l} \rightarrow \mathbb{F}_{q^l}$ denote the Frobenius map defined by $z \mapsto z^q$. By abuse of notation, we also use τ to represent its entry-wise extension to vectors and matrices. Let $\lambda \in \mathbb{F}_{q^l}$ be an eigenvalue of S and $\xi \in \mathbb{F}_{q^l}^l$ be the eigenvector of S corresponding to eigenvalue λ . One can check that $\tau^i(\xi)$ is the eigenvector corresponding to $\tau^i(\lambda)$ and L is invertible due to the fact that the characteristic polynomial of S is irreducible. Let $L = [\xi \ \tau(\xi) \ \cdots \ \tau^{l-1}(\xi)] \in \mathbb{F}_{q^l}^{l \times l}$, then one can have

$$L^{-1} \cdot S \cdot L = \begin{bmatrix} \lambda & & & \\ & \lambda^q & & \\ & & \ddots & \\ & & & \lambda^{q^{l-1}} \end{bmatrix} = \begin{bmatrix} \lambda & & & \\ & \tau(\lambda) & & \\ & & \ddots & \\ & & & \tau^{l-1}(\lambda) \end{bmatrix}. \quad (2.4)$$

Let $\mathbf{U}$ be the commutation matrix corresponding to index (n, l) , i.e.,

$$\mathbf{U} = \begin{bmatrix} E_{11} & E_{21} & \cdots & E_{l1} \\ E_{12} & E_{22} & \cdots & E_{l2} \\ \vdots & \vdots & \ddots & \vdots \\ E_{1l} & E_{2l} & \cdots & E_{ll} \end{bmatrix} \quad (2.5)$$

where E_{ij} is the matrix of size $l \times n$ whose (i, j) -entry is one and other entries are zero. In addition, we define the following notation

$$\bar{\mathbf{S}} = \mathbf{U}^\top \cdot (\mathbf{L}^{-1} \mathbf{S} \mathbf{L})^{\otimes n} \cdot \mathbf{U} = \mathbf{U}^\top (\mathbf{L}^{\otimes n})^{-1} \mathbf{S}^{\otimes n} \mathbf{L}^{\otimes n} \mathbf{U}. \quad (2.6)$$

Note that $\mathbf{U}^{-1} = \mathbf{U}^\top$. Thus, we have

$$\bar{\mathbf{S}}^a = (\mathbf{U}^\top \cdot (\mathbf{L}^{-1} \mathbf{S} \mathbf{L})^{\otimes n} \cdot \mathbf{U})^a = \mathbf{U}^\top (\mathbf{L}^{\otimes n})^{-1} (\mathbf{S}^a)^{\otimes n} \mathbf{L}^{\otimes n} \mathbf{U}. \quad (2.7)$$

Given the SNOVA public key $\mathbf{P}_1, \dots, \mathbf{P}_m \in \mathbb{F}_q^{ln \times ln}$, the following Theorem 1 demonstrates how to lift SNOVA to the extension field $\mathbb{F}_q^l$.

Theorem 1 ([7, Theorem 1]). *Let $\bar{\mathbf{T}} = \mathbf{U}^\top \cdot (\mathbf{L}^{\otimes n})^{-1} \cdot \mathbf{T} \cdot \mathbf{L}^{\otimes n} \cdot \mathbf{U}$. For $k \in \{1, \dots, m\}$, let $\bar{\mathbf{F}}_k = \mathbf{U}^\top \cdot (\mathbf{L}^{\otimes n})^\top \cdot \mathbf{F}_k \cdot \mathbf{L}^{\otimes n} \cdot \mathbf{U}$ and $\bar{\mathbf{P}}_k = \mathbf{U}^\top \cdot (\mathbf{L}^{\otimes n})^\top \cdot \mathbf{P}_k \cdot \mathbf{L}^{\otimes n} \cdot \mathbf{U}$. For each $i, j \in \{1, \dots, l\}$, we define $\bar{F}_{k,(i,j)} \in \mathbb{F}_{q^l}^{n \times n}$ and $\bar{P}_{k,(i,j)} \in \mathbb{F}_{q^l}^{n \times n}$ by the (i, j) -th submatrices of $\bar{\mathbf{F}}_k$ and $\bar{\mathbf{P}}_k$, respectively. Then, we have*

(i) For all k, i, j , the lower right o-by-o submatrix of $\bar{F}_{k,(i,j)}$ is the zero matrix, i.e., $\bar{F}_{k,(i,j)}$ is a UOV-type matrix, and we have

$$\tau(\bar{F}_{k,(i,j)}) = \bar{F}_{k,((i+1) \bmod l, (j+1) \bmod l)}. \quad (2.8)$$

More precisely, we have

$$\bar{\mathbf{F}}_k = \begin{bmatrix} \bar{F}_{k,(1,1)} & \bar{F}_{k,(1,2)} & \cdots & \bar{F}_{k,(1,l)} \\ \tau(\bar{F}_{k,(1,l)}) & \tau(\bar{F}_{k,(1,1)}) & \cdots & \tau(\bar{F}_{k,(1,l-1)}) \\ \vdots & \vdots & \ddots & \vdots \\ \tau^{l-1}(\bar{F}_{k,(1,2)}) & \tau^{l-1}(\bar{F}_{k,(1,3)}) & \cdots & \tau^{l-1}(\bar{F}_{k,(1,1)}) \end{bmatrix}, \quad (2.9)$$

(ii) For the matrix $\bar{\mathbf{T}} = \mathbf{U}^\top \cdot (\mathbf{L}^{\otimes n})^{-1} \cdot \mathbf{T} \cdot \mathbf{L}^{\otimes n} \cdot \mathbf{U}$, we have

$$\bar{\mathbf{T}} = \begin{bmatrix} \bar{T}_1 & & & \\ & \tau(\bar{T}_1) & & \\ & & \ddots & \\ & & & \tau^{l-1}(\bar{T}_1) \end{bmatrix}, \quad (2.10)$$

where $\bar{T}_1 \in \mathbb{F}_{q^l}^{n \times n}$ is of the form

$$\begin{bmatrix} I_{v \times v} & *_{v \times o} \\ 0 & I_{o \times o} \end{bmatrix}. \quad (2.11)$$

(iii) For all k, i, j , we have

$$\tau(\bar{P}_{k,(i,j)}) = \bar{P}_{k,((i+1) \bmod l, (j+1) \bmod l)}, \quad (2.12)$$

$$\bar{P}_{k,(i,j)} = (\tau^{i-1}(\bar{T}_1))^\top \cdot \bar{F}_{k,(i,j)} \cdot \tau^{j-1}(\bar{T}_1). \quad (2.13)$$

2.3 Inverse Problem of Lifting

In the above Section 2.2, we reviewed how a SNOVA public key $\mathbf{P}_1, \dots, \mathbf{P}_m \in \mathbb{F}_q^{ln \times ln}$ can be transformed into a set of public keys over an extension field. The natural inverse question then arises: What kind of matrices should be given, and how can we construct a suitable set of matrices so that they can be transformed back into a SNOVA public key over the base field $\mathbb{F}_q$?

We start with the following result in [11].

Lemma 1 ([11, Theorem B.1]). *Let q be a power of a prime, and let $\mathbb{F}_q$ be the finite field of order q . For any monic irreducible polynomial $f(x) \in \mathbb{F}_q[x]$ of degree l , there exists a symmetric matrix $S \in \mathbb{F}_q^{l \times l}$ whose characteristic polynomial is $f(x)$.*

Therefore, given an extension field $\mathbb{F}_{q^l}$, by above Lemma 1, we always can find a symmetric matrix $S \in \mathbb{F}_q^{l \times l}$ related to $\mathbb{F}_{q^l}$. Note that, every element in the extension field $\mathbb{F}_{q^l}$ can be represented in the form of element in $\mathbb{F}_q[S]$. Note that, by counting dimension, one can observe that the map $\varphi : \mathbb{F}_q^{ln \times ln} \rightarrow \mathcal{H}^{n \times n}$ defined by $\mathbf{G} \mapsto (\mathbf{L}^{\otimes n})^\top \cdot \mathbf{G} \cdot \mathbf{L}^{\otimes n}$ is a linear isomorphism where $\mathcal{H}$ denotes the space consisting of the matrices of the form

$$\begin{bmatrix} h_1 & h_2 & \cdots & h_l \\ \tau(h_1) & \tau(h_2) & \cdots & \tau(h_{l-1}) \\ \vdots & \vdots & \ddots & \vdots \\ \tau^{l-1}(h_2) & \tau^{l-1}(h_3) & \cdots & \tau^{l-1}(h_1) \end{bmatrix} \quad (2.14)$$

and $h_1, h_2, \dots, h_l \in \mathbb{F}_{q^l}$.

Next, we first choose $\bar{F}_{k,(1,1)}, \dots, \bar{F}_{k,(1,l)} \in \mathbb{F}_{q^l}^{n \times n}$ so that in each $\bar{F}_{k,(1,j)}$ the lower right o -by- o submatrix is zero matrix and choose a $\bar{T}_1 \in \mathbb{F}_{q^l}^{n \times n}$ to be of UOV-type. Then, we can use the Frobenius map $\tau : \mathbb{F}_{q^l} \rightarrow \mathbb{F}_{q^l}$ to construct $\bar{\mathbf{F}}_k$ and $\bar{\mathbf{T}}$ of the form in Equation (2.9) and (2.10). Therefore, for the matrices $\bar{\mathbf{F}}_1, \dots, \bar{\mathbf{F}}_m$, we define

$$\mathbf{F}_k = \varphi^{-1}((\mathbf{U}^\top)^{-1} \bar{\mathbf{F}}_k \mathbf{U}^{-1}). \quad (2.15)$$

Then, the matrices $\mathbf{F}_1, \dots, \mathbf{F}_m$ are the desired SNOVA public key over based field $\mathbb{F}_q$. Similarly, we can use the matrix $\bar{\mathbf{T}}$ to find the corresponding SNOVA private key $\mathbf{T}$ over based field $\mathbb{F}_q$.

Relationship Between SNOVA and Lifted SNOVA. Note that once the oil space of the lifted SNOVA is found, it can then be used to recover the oil space of SNOVA over the base field. On the other hand, the discussion in Section 2.2 and Section 2.3 shows that there exists a natural correspondence between SNOVA and lifted SNOVA due to the effect of isomorphism φ .

The above two facts seem to suggest that we should instead consider the wedge attack over the extension field $\mathbb{F}_{q^l}$. In fact, by considering the wedge attack over extension field $\mathbb{F}_{q^l}$, we have managed to resolve the difficulties in clearly formulating the wedge attack against SNOVA over the base field $\mathbb{F}_q$ and correctly predict the dimension of the kernel (and hence the rank) of the wedge map.

3 Wedge Attack

This section describes the construction of the wedge attack against SNOVA and its lifted version.

3.1 Wedge Attack on SNOVA

Lars Ran was the first to consider the wedge attack on SNOVA in his work [13]. However, as he pointed out in his paper, the approach for the prediction of the kernel dimension proposed in the paper do not directly apply to SNOVA. This leaves the precise mechanism of how the wedge attack operates against SNOVA somewhat unclear. The wedge attack on SNOVA proposed by Le *et al.* [9] extended Rans wedge attack on UOV [13] to the structured SNOVA setting by exploiting the $\mathbb{F}_q[S]$ -module constraints. In [9], the authors attempted to provide an estimate of the kernel dimension of the wedge map in SNOVA through two different approaches. Nevertheless, we believe that their method is neither complete nor sufficiently transparent.

In this subsection, we present a clear characterization of the wedge attack on SNOVA. In the subsequent sections, we further introduce a generating function and show that it allows us to accurately predict the kernel dimension of the wedge map, thereby resolving the the issues left in previous analyses [13,9].

We review their construction and adapt it to our notation.

Let $q = 2^k$ be a power of 2, and let $\mathbf{P}_1, \dots, \mathbf{P}_m \in \mathbb{F}_q^{ln \times ln}$ be the public key of a (v, o, q, l, m) SNOVA scheme. Here, we define the space $\mathcal{O}_1$ by

$$\mathcal{O}_1 = \{\mathbf{x} = (x_1, \dots, x_{ln}) \in \mathbb{F}_q^{ln} : x_1 = \dots = x_{lv} = 0\} \quad (3.1)$$

and we define $\mathcal{O} = \mathbf{T}^{-1}(\mathcal{O}_1)$.

A family of bilinear forms. Because the oil space of SNOVA is block-wise restricted over the subring $\mathbb{F}_q[S] \subseteq \mathbb{F}_q^{l \times l}$, it is natural (and standard in SNOVA cryptanalysis) to enlarge the set of public bilinear forms as follows. Since $\mathbf{T} \in (\mathbb{F}_q[S])^{n \times n}$, the private key $\mathbf{T}$ satisfying the following

$$\mathbf{T} \cdot \mathbf{S}^{\otimes n} = \mathbf{S}^{\otimes n} \cdot \mathbf{T}. \quad (3.2)$$

Thus

$$\mathbf{S}^{\otimes n} \mathcal{O} = \mathbf{S}^{\otimes n} \mathbf{T}^{-1}(\mathcal{O}_1) = \mathbf{T}^{-1} \mathbf{S}^{\otimes n}(\mathcal{O}_1) = \mathbf{T}^{-1}(\mathcal{O}_1) = \mathcal{O}. \quad (3.3)$$

Therefore, for each oil vector $\mathbf{x} \in \mathcal{O}$, we have

$$\mathbf{S}^{\otimes n} \cdot \mathbf{x}, \dots, (\mathbf{S}^{\otimes n})^{l-1} \cdot \mathbf{x} \in \mathcal{O}. \quad (3.4)$$

For $a, b \in \{0, \dots, l-1\}$, we define

$$\mathbf{P}_k^{(a,b)} := (\mathbf{S}^a)^{\otimes n} \mathbf{P}_k (\mathbf{S}^b)^{\otimes n} \in \mathbb{F}_q^{ln \times ln}. \quad (3.5)$$

Then, for all $\mathbf{x}, \mathbf{y} \in \mathcal{O}$ and for $a, b \in \{0, \dots, l-1\}$, we then have

$$\mathbf{x}^\top \cdot \mathbf{P}_k^{(a,b)} \cdot \mathbf{y} = \mathbf{x}^\top \cdot (\mathbf{S}^{\otimes n})^a \mathbf{P}_k (\mathbf{S}^{\otimes n})^b \cdot \mathbf{y} = 0 \quad (3.6)$$

where $k \in \{1, \dots, m\}$. Therefore, all forms $\{\mathbf{P}_k^{(a,b)}\}_{k,a,b}$ vanish on $\mathcal{O} \times \mathcal{O}$. Thus, starting from only m public matrices, an adversary may work with an explicit family of ml^2 bilinear forms.

From bilinear forms to alternating 2-forms. For each derived public matrix $\mathbf{P}_k^{(a,b)} \in \mathbb{F}_q^{ln \times ln}$, define the bilinear form

$$B_k^{(a,b)}(\mathbf{x}, \mathbf{y}) := \mathbf{x}^\top \mathbf{P}_k^{(a,b)} \mathbf{y}, \quad \mathbf{x}, \mathbf{y} \in \mathbb{F}_q^{ln}, \quad (3.7)$$

and its alternating part (in characteristic 2) by

$$Q_k^{(a,b)}(\mathbf{x}, \mathbf{y}) := B_k^{(a,b)}(\mathbf{x}, \mathbf{y}) + B_k^{(a,b)}(\mathbf{y}, \mathbf{x}) = \mathbf{x}^\top \left(\mathbf{P}_k^{(a,b)} + (\mathbf{P}_k^{(a,b)})^\top \right) \mathbf{y}. \quad (3.8)$$

We identify $Q_k^{(a,b)}$ with the corresponding exterior 2-form $\widehat{Q}_k^{(a,b)} \in \bigwedge^2 \mathbb{F}_q^{ln}$. Since $B_k^{(a,b)}(\mathbf{x}, \mathbf{y}) = 0$ for all $\mathbf{x}, \mathbf{y} \in \mathcal{O}$, we also have $Q_k^{(a,b)}(\mathbf{x}, \mathbf{y}) = 0$ for all $\mathbf{x}, \mathbf{y} \in \mathcal{O}$. Let $\mathcal{A} := \mathcal{O}^\perp$ be the annihilator space of $\mathcal{O}$, so $\dim(\mathcal{A}) = lv$. Fix a basis $\mathbf{a}_1, \dots, \mathbf{a}_{lv}$ of $\mathcal{A}$, and set

$$\widehat{\mathcal{A}} := \mathbf{a}_1 \wedge \dots \wedge \mathbf{a}_{lv} \in \bigwedge^{lv} \mathbb{F}_q^{ln}. \quad (3.9)$$

Lemma 2 ([13, Lemma 2] and [9, Lemma 1]). *Let $\mathcal{O} \subset \mathbb{F}_q^{ln}$ and $\mathcal{A} := \mathcal{O}^\perp$. Let $\mathbf{a}_1, \dots, \mathbf{a}_{lv}$ be a basis of $\mathcal{A}$. If $\widehat{Q} \in \bigwedge^2 \mathbb{F}_q^{ln}$ is an alternating 2-form whose associated polar form satisfies $Q(\mathbf{x}, \mathbf{y}) = 0$ for all $\mathbf{x}, \mathbf{y} \in \mathcal{O}$, then there exist vectors $\mathbf{x}_1, \dots, \mathbf{x}_{lv} \in \mathbb{F}_q^{ln}$ such that*

$$\widehat{Q} = \sum_{j=1}^{lv} \mathbf{a}_j \wedge \mathbf{x}_j. \quad (3.10)$$

Applying Lemma 2 to $\widehat{Q}_k^{(a,b)}$, we obtain vectors $\mathbf{x}_1, \dots, \mathbf{x}_{lv} \in \mathbb{F}_q^{ln}$ such that

$$\widehat{Q}_k^{(a,b)} = \sum_{j=1}^{lv} \mathbf{a}_j \wedge \mathbf{x}_j. \quad (3.11)$$

Wedge-multiplying by $\widehat{\mathcal{A}} := \mathbf{a}_1 \wedge \dots \wedge \mathbf{a}_{lv}$ yields

$$\widehat{\mathcal{A}} \wedge \widehat{Q}_k^{(a,b)} = \mathbf{a}_1 \wedge \dots \wedge \mathbf{a}_{lv} \wedge \left(\sum_{j=1}^{lv} \mathbf{a}_j \wedge \mathbf{x}_j \right) = 0, \quad (3.12)$$

for all $k \in \{1, \dots, m\}$ and for all $a, b \in \{0, \dots, l-1\}$.

The wedge map. Motivated by the relations $\widehat{\mathcal{A}} \wedge \widehat{Q}_k^{(a,b)} = 0$, define the wedge map for SNOVA by

$$\mathcal{M}_{\text{SNOVA}} : \bigwedge^{lv} \mathbb{F}_q^{ln} \rightarrow \left(\bigwedge^{lv+2} \mathbb{F}_q^{ln} \right)^{ml^2}, \quad \omega \mapsto \left(\omega \wedge \widehat{Q}_k^{(a,b)} \right)_{\substack{k \in \{1, \dots, m\} \\ a, b \in \{0, \dots, l-1\}}} . \quad (3.13)$$

By construction, the (unknown) blade $\hat{\mathcal{A}}$ lies in the kernel of $\mathcal{M}_{\text{SNOVA}}$. Hence the wedge attack reduces key recovery to a single structured linear-algebra problem: compute $\ker(\mathcal{M}_{\text{SNOVA}})$ (typically via sparse linear algebra). If this kernel is one-dimensional, then $\hat{\mathcal{A}}$ is determined up to scaling and one can reconstruct $\mathcal{A}$ (and hence the oil space $\mathcal{O}$) from $\hat{\mathcal{A}}$ by standard Plücker-coordinate techniques.

A structured domain for $\mathcal{M}_{\text{SNOVA}}$. To make the search space more explicit, view the annihilator space $\mathcal{A} = \mathcal{O}^\perp$ as an $\mathbb{F}_q[S]$ -submodule of $(\mathbb{F}_q[S])^n$ of rank v . Fix an $\mathbb{F}_q[S]$ -basis of $\mathcal{A}$ and write it as a matrix

$$\mathbf{A} \in (\mathbb{F}_q[S])^{n \times v}, \quad (3.14)$$

whose columns form such a basis. Let $\iota : \mathbb{F}_q[S] \hookrightarrow \mathbb{F}_q^{l \times l}$ be the canonical injection and extend it entry-wise to matrices. Thus,

$$\iota(\mathbf{A}) \in \mathbb{F}_q^{ln \times lv}. \quad (3.15)$$

The Plücker vector $\hat{\mathcal{A}} \in \bigwedge^{lv} \mathbb{F}_q^{ln}$ can be identified with the exterior product of the columns of $\iota(\mathbf{A})$; equivalently, the Plücker coordinates of $\hat{\mathcal{A}}$ are precisely the $lv \times lv$ maximal minors of $\iota(\mathbf{A})$.

Proposition 1 of [9] implies that these maximal minors are not independent: the set of all Plücker coordinates of $\hat{\mathcal{A}}$ lies in an $\mathbb{F}_q$ -subspace

$$\mathcal{D} \subseteq \bigwedge^{lv} \mathbb{F}_q^{ln} \quad (3.16)$$

of dimension $\dim_{\mathbb{F}_q}(\mathcal{D}) = \binom{n}{v}^l$. In particular, although $\mathcal{M}_{\text{SNOVA}}$ is defined on the full space $\bigwedge^{lv} \mathbb{F}_q^{ln}$ of dimension $\binom{ln}{lv}$, the unknown kernel element we seek satisfies $\hat{\mathcal{A}} \in \mathcal{D}$. Hence, for the purpose of recovering $\hat{\mathcal{A}}$, it is sufficient to restrict the wedge map to $\mathcal{D}$, i.e., to work with

$$\mathcal{M}_{\text{SNOVA}}|_{\mathcal{D}} : \mathcal{D} \longrightarrow \left(\bigwedge^{lv+2} \mathbb{F}_q^{ln} \right)^{ml^2}. \quad (3.17)$$

We will revisit this restriction in the next subsection from the lifted SNOVA perspective, where the structure and the dimension of $\mathcal{D}$ become more explicit.

3.2 Wedge Attack on the Lifted SNOVA

We now pass to a lifted (base-changed) viewpoint. The key technical point is that the rank of the wedge map is invariant under base change.

Theorem 2. *Let F be a field and K/F a field extension. Let V, W be F -vector spaces and let $f : V \rightarrow W$ be an F -linear map. Define the base-changed map $f_K := f \otimes \text{id}_K : V \otimes_F K \rightarrow W \otimes_F K$. Then*

$$\dim_F(\text{im}(f)) = \dim_K(\text{im}(f_K)). \quad (3.18)$$

In particular, the rank ($= \dim(\text{im}(f))$) is preserved under base change.

Proof. Consider the short exact sequence of F -vector spaces

$$0 \longrightarrow \ker(f) \longrightarrow V \xrightarrow{f} \operatorname{im}(f) \longrightarrow 0. \quad (3.19)$$

Since K is a field extension of F , the F -module K is an F -vector space and hence free. Therefore K is flat over F (see [5, Chapter 10, Corollary 42]). Therefore, tensoring with K preserves exactness, and we obtain an exact sequence

$$0 \longrightarrow \ker(f) \otimes_F K \longrightarrow V \otimes_F K \xrightarrow{f \otimes \operatorname{id}_K} \operatorname{im}(f) \otimes_F K \longrightarrow 0. \quad (3.20)$$

In particular,

$$\operatorname{im}(f_K) = \operatorname{im}(f \otimes \operatorname{id}_K) \cong \operatorname{im}(f) \otimes_F K. \quad (3.21)$$

For any F -vector space U , we have

$$\dim_K(U \otimes_F K) = \dim_F(U). \quad (3.22)$$

Applying this to $U = \operatorname{im}(f)$, we conclude

$$\dim_K(\operatorname{im}(f_K)) = \dim_K(\operatorname{im}(f) \otimes_F K) = \dim_F(\operatorname{im}(f)). \quad (3.23)$$

□

Applying Theorem 2 to the lifting setting, we fix the extension field $K := \mathbb{F}_{q^l}$ and set $\bar{\mathcal{M}}_{\text{SNOVA}} := \mathcal{M}_{\text{SNOVA}} \otimes \operatorname{id}_{\mathbb{F}_{q^l}}$. Then

$$\operatorname{rank}(\mathcal{M}_{\text{SNOVA}}) = \operatorname{rank}(\bar{\mathcal{M}}_{\text{SNOVA}}). \quad (3.24)$$

In the lifted SNOVA model, we pass to $\mathbb{F}_{q^l}$ and replace the original SNOVA data $(\mathbf{T}, \mathbf{P}_1, \dots, \mathbf{P}_m)$ by an equivalent lifted representation $(\bar{\mathbf{T}}, \bar{\mathbf{P}}_1, \dots, \bar{\mathbf{P}}_m)$, obtained via an explicit invertible $\mathbb{F}_{q^l}$ -linear change of basis. Consequently, after base change the wedge map

$$\bar{\mathcal{M}}_{\text{SNOVA}} : \bigwedge^{lv} \mathbb{F}_{q^l}^{ln} \longrightarrow \left(\bigwedge^{lv+2} \mathbb{F}_{q^l}^{ln} \right)^{ml^2} \quad (3.25)$$

may be evaluated using $\bar{\mathbf{T}}$ and $\bar{\mathbf{P}}_k$ in place of $\mathbf{T}$ and $\mathbf{P}_k$, since conjugation by an invertible $\mathbb{F}_{q^l}$ -linear transformation preserves the rank. This viewpoint is particularly convenient in practice. The lifted representation makes the underlying $\mathbb{F}_q[S]$ -module structure explicit, and typically yields a sparser and better-structured linear system for the wedge relations defining $\bar{\mathcal{M}}_{\text{SNOVA}}$. As a result, the behavior of the wedge attack on SNOVA becomes easier to analyze.

The lifted SNOVA viewpoint makes the structure underlying the reduced domain more explicit. After extending scalars to $\mathbb{F}_{q^l}$, Theorem 1 provides an explicit conjugation which transforms the private linear map into a block-diagonal form:

$$\bar{\mathbf{T}} = \begin{bmatrix} \bar{T}_1 & & & \\ & \tau(\bar{T}_1) & & \\ & & \ddots & \\ & & & \tau^{l-1}(\bar{T}_1) \end{bmatrix}, \quad \bar{T}_1 = \begin{bmatrix} I_{v \times v} & * \\ 0 & I_{o \times o} \end{bmatrix} \in \mathbb{F}_{q^l}^{n \times n}. \quad (3.26)$$

In particular, over $\mathbb{F}_{q^l}$ the $\mathbb{F}_q[S]$ -module constraints split into l Frobenius conjugate components. If $\mathbf{A} \in (\mathbb{F}_q[S])^{n \times v}$ is an $\mathbb{F}_q[S]$ -basis matrix for the annihilator module $\mathcal{A}$, then after expansion via canonical injection ι and applying the same change of basis as in Theorem 1, the matrix $\iota(\mathbf{A})$ becomes equivalent (over $\mathbb{F}_{q^l}$) to a block structure consisting of l independent $n \times v$ components. Consequently, any maximal $lv \times lv$ minor which remains nonzero after lifting is obtained by choosing v rows from each of the l components. This yields exactly $\binom{n}{v}^l$ degrees of freedom among the Plücker (minor) coordinates, which is precisely the content of Proposition 1 of [9].

This splitting also clarifies the appropriate domain restriction for the wedge attack over $\mathbb{F}_{q^l}$. The base-changed wedge map $\bar{\mathcal{M}}_{\text{SNOVA}}$ is $\mathbb{F}_{q^l}$ -linear with domain $\bigwedge^{lv} \mathbb{F}_{q^l}^{ln}$, whose ambient dimension is $\binom{ln}{lv}$. However, the kernel element of interest is the Plücker vector $\hat{\mathcal{A}}$, and it is supported only on those Plücker coordinates arising from annihilator bases, i.e., on the maximal minors described above. Hence one should restrict $\bar{\mathcal{M}}_{\text{SNOVA}}$ to the scalar extension of the structured subspace $\mathcal{D} \subseteq \bigwedge^{lv} \mathbb{F}_q^{ln}$,

$$\bar{\mathcal{D}} := \mathcal{D} \otimes_{\mathbb{F}_q} \mathbb{F}_{q^l} \subseteq \bigwedge^{lv} \mathbb{F}_{q^l}^{ln}, \quad (3.27)$$

where $\dim_{\mathbb{F}_{q^l}}(\bar{\mathcal{D}}) = \binom{n}{v}^l$ by [9, Proposition 1]. In practice, working directly with the lifted keys $(\bar{\mathbf{T}}, \bar{\mathbf{P}}_k)$ makes this restriction explicit, and the resulting linear system for the wedge relations is substantially smaller and sparser. This is the main reason why computing $\text{rank}(\bar{\mathcal{M}}_{\text{SNOVA}})$ is considerably simpler and more efficient experimentally, and it also provides the starting point for our rank prediction in the next section.

3.3 Shrinking the Co-Domain of the Wedge Map on the Lifted SNOVA

As established by Le *et al.* [9], the wedge map built from ml^2 alternating forms and ml chosen alternating 2-forms possess identical ranks. In this work, we utilize the wedge map on lifted SNOVA to explicitly prove that these constructions share the same kernel, thereby confirming their same rank. In section 5, we explicitly write out the form of $\bar{\mathcal{D}}$ and provide an explicit expression for $\bar{\mathcal{M}}$ in the extension field.

To show that the wedge map for SNOVA by

$$\mathcal{M}_{\text{SNOVA}}|_{\mathcal{D}} : \mathcal{D} \rightarrow \left(\bigwedge^{lv+2} \mathbb{F}_q^{ln} \right)^{ml^2}, \quad \omega \mapsto \left(\omega \wedge \hat{Q}_k^{(a,b)} \right)_{\substack{k \in \{1, \dots, m\} \\ a, b \in \{0, \dots, l-1\}}} \quad (3.28)$$

and the wedge map for SNOVA with co-domain shrinking by

$$\mathcal{M}_{\text{SNOVA}}|_{\mathcal{D}} : \mathcal{D} \rightarrow \left(\bigwedge^{lv+2} \mathbb{F}_q^{ln} \right)^{ml}, \quad \omega \mapsto \left(\omega \wedge \hat{Q}_k^{(0,b)} \right)_{\substack{k \in \{1, \dots, m\} \\ b \in \{0, \dots, l-1\}}} \quad (3.29)$$

having the same rank, it is equivalent to showing that the wedge map on the lifted SNOVA and the wedge map on the lifted SNOVA with co-domain shrinking having the same rank. Therefore, we complete the proof if we can show that the wedge map on the lifted SNOVA and the wedge map on the lifted SNOVA with co-domain shrinking having the same kernel.

Let $\omega \in \bar{\mathcal{D}}$ be represented by

$$\omega = \sum \mathbf{X}_{i_1, i_2, \dots, i_{lv}} \mathbf{e}_{i_1} \wedge \dots \wedge \mathbf{e}_{i_{lv}} \quad (3.30)$$

where

$$i_1, i_2, \dots, i_v \in \{1, 2, \dots, n\}, \quad (3.31)$$

$$i_{v+1}, i_{v+2}, \dots, i_{2v} \in \{n+1, n+2, \dots, 2n\}, \dots, \quad (3.32)$$

$$i_{(l-1)v+1}, i_{(l-1)v+2}, \dots, i_{lv} \in \{(l-1)n+1, (l-1)n+2, \dots, ln\}. \quad (3.33)$$

Suppose that $\omega \wedge \hat{Q}_k^{(0,b)} = \mathbf{0}$ for any $k \in \{1, \dots, m\}$ and $b \in \{0, \dots, l-1\}$. Since a zero form after the coordinate change $\mathbf{e}_i \mapsto \bar{\mathbf{S}}^a \mathbf{e}_i$ is still a zero form,

$$\left(\sum \mathbf{X}_{i_1, i_2, \dots, i_{lv}} \mathbf{e}_{i_1} \wedge \dots \wedge \mathbf{e}_{i_{lv}} \right) \wedge \hat{Q}_k^{(0,b-a)} = \omega \wedge \hat{Q}_k^{(0,b-a)} = \mathbf{0} \quad (3.34)$$

implies

$$\mathbf{0} = \sum \mathbf{X}_{i_1, i_2, \dots, i_{lv}} (\bar{\mathbf{S}}^a \mathbf{e}_{i_1}) \wedge \dots \wedge (\bar{\mathbf{S}}^a \mathbf{e}_{i_{lv}}) \wedge (\bar{\mathbf{S}}^a \hat{Q}_k^{(0,b-a)} \bar{\mathbf{S}}^a). \quad (3.35)$$

That is,

$$\mathbf{0} = \lambda^{av+avq+\dots+avq^{l-1}} \sum \mathbf{X}_{i_1, i_2, \dots, i_{lv}} \mathbf{e}_{i_1} \wedge \dots \wedge \mathbf{e}_{i_{lv}} \wedge \hat{Q}_k^{(a,b)}. \quad (3.36)$$

Hence, $\omega \wedge \hat{Q}_k^{(a,b)} = \mathbf{0}$. Therefore, a wedge map with co-domain shrinking is the preferred option for the wedge attack, since it enhances performance while retaining all information.

3.4 Experiment Results

Building on the arguments in the previous subsections, we have shown how to properly formulate the wedge attack against SNOVA over the base field $\mathbb{F}_q$, as well as the wedge attack against lifted SNOVA over the extension field $\mathbb{F}_{q^l}$. Moreover, we proved that once these two attacks are correctly formulated, they are in fact equivalent. We also demonstrated that considering ml^2 2-forms instead of ml 2-forms does not affect the estimation of the actual rank or the kernel dimension of the wedge map. In this subsection, we present experimental data obtained from our Magma implementations of the wedge attack against lifted SNOVA over the extension field $\mathbb{F}_{q^l}$ in Table 1. From the reported results, we observe that the experimental evidence supports the theoretical arguments established in the previous sections.

Table 1. Table of the rank of the wedge map $\text{rank}(\bar{\mathcal{M}}_{\text{SNOVA}})$ obtained by our Magma implementation together with m , ml^2 and ml 2-forms respectively. Note that the number with “*” in the table is different from the reported data by Le *et al.* [9]. We believe that the number they reported might be a typo.

(v, o, q, l, m, n)	rank of wedge map with		
	m 2-forms	ml^2 2-forms	ml 2-forms
(3, 2, 16, 2, 1, 5)	43	66	66
(4, 2, 16, 2, 1, 6)	64	98	98
(4, 2, 16, 2, 2, 6)	125	190	190
(4, 3, 16, 2, 1, 7)	793*	1098*	1098*
(5, 2, 16, 2, 1, 7)	89	136	136
(2, 1, 16, 3, 1, 3)	9	18	18
(3, 1, 16, 3, 1, 4)	12	24	24
(2, 1, 16, 4, 1, 3)	51	80	80

4 Application to SNOVA Parameter Sets

4.1 Generating Function for the Dimension of the Kernel

In this subsection, we explain how the framework of the wedge attack against lifted SNOVA provides insight into the correct estimation formula for the dimension of the kernel of the wedge map. It is worth noting that the generating function we propose in this section is fully consistent with our experimental results in the Section 3.4.

We begin with the theorem proposed by Lars Ran in [13].

Theorem 3 ([13, Proposition 2]). *Let $(p_1, \dots, p_m)$ be (v, o, m) -UOV public maps considered as elements of the polynomial ring*

$$\mathbb{F}_q[x_1, \dots, x_n] / \langle x_1^2, \dots, x_n^2 \rangle$$

such that $(p_1, \dots, p_m)$ is Boolean semi-regular. Let $q_1, \dots, q_m$ be UOV polar forms. Then the linear map

$$\mathcal{M} : \bigwedge^v \mathbb{F}_q^n \rightarrow \left(\bigwedge^{v+2} \mathbb{F}_q^n \right)^m \quad (4.1)$$

$$\alpha \mapsto (\alpha \wedge q_1, \dots, \alpha \wedge q_m) \quad (4.2)$$

has a kernel of dimension

$$T_{m,n}[t^o] \quad (4.3)$$

where

$$T_{m,n} = \frac{(1+t)^n}{(1+t^2)^m}. \quad (4.4)$$

First, from the generating function $T_{m,n}$ we can observe that the coefficient

$$(1+t)^n [t^o] = \binom{n}{o} = \binom{n}{v} = \dim \left(\bigwedge^v \mathbb{F}_q^n \right). \quad (4.5)$$

Thus, the numerator in the generating function gives the dimension of the domain of the wedge map $\mathcal{M}$. In the case of SNOVA, by observing that the $\dim_{\mathbb{F}_{q^l}}(\bar{\mathcal{D}}) = \binom{n}{v}^l$. The block-diagonal structure of the matrix $\bar{T}$ suggests that the numerator of the generating function should be

$$(1+t_1)^n \cdots (1+t_l)^n. \quad (4.6)$$

Similar to the case of Theorem 3, we could notice that each term $(1+t_i)^n$ will contribute a factor $\binom{n}{v}$ and hence in total $\binom{n}{v}^l = \dim_{\mathbb{F}_{q^l}}(\bar{\mathcal{D}})$. According to the dimension counting, our guess on the numerator seems reasonable.

Next, let us observe the structure of matrices $\bar{\mathbf{P}}_1, \dots, \bar{\mathbf{P}}_m$. By Theorem 1, for each $k \in \{1, \dots, m\}$, we have

$$\bar{\mathbf{P}}_k = \begin{bmatrix} \bar{P}_{k,(1,1)} & \bar{P}_{k,(1,2)} & \cdots & \bar{P}_{k,(1,l)} \\ \tau(\bar{P}_{k,(1,l)}) & \tau(\bar{P}_{k,(1,1)}) & \cdots & \tau(\bar{P}_{k,(1,l-1)}) \\ \vdots & \vdots & \ddots & \vdots \\ \tau^{l-1}(\bar{P}_{k,(1,2)}) & \tau^{l-1}(\bar{P}_{k,(1,3)}) & \cdots & \tau^{l-1}(\bar{P}_{k,(1,1)}) \end{bmatrix}. \quad (4.7)$$

Because we already know that $\binom{n}{v}^l = \dim_{\mathbb{F}_{q^l}}(\bar{\mathcal{D}})$, i.e., the maximal minors provided by the diagonal blocks $\tau(\bar{T}_1), \dots, \tau^{l-1}(\bar{T}_1)$ of the matrix $\bar{\mathbf{T}}$ are linearly independent. Therefore, the case of $i = j$, the diagonal blocks

$$\bar{P}_{k,(i,i)} = (\tau^{i-1}(\bar{T}_1))^\top \cdot \bar{F}_{k,(i,i)} \cdot \tau^{i-1}(\bar{T}_1) \quad (4.8)$$

suggest that there should have a factor $(1+t_i^2)^m$ in the denominator of the generating function. This term corresponds to the complete quadratic term in the framework of wedge attack. On the other hand, the case of $i \neq j$, the remaining blocks

$$\bar{P}_{k,(i,j)} = (\tau^{i-1}(\bar{T}_1))^\top \cdot \bar{F}_{k,(i,j)} \cdot \tau^{j-1}(\bar{T}_1) \quad (4.9)$$

and

$$\bar{P}_{k,(j,i)} = (\tau^{j-1}(\bar{T}_1))^\top \cdot \bar{F}_{k,(j,i)} \cdot \tau^{i-1}(\bar{T}_1) \quad (4.10)$$

suggest that there should have a factor $(1+t_i t_j)^{2m}$ in the denominator of the generating function. Based on the above observations and insights, we obtained the corresponding generating function

$$H_{m,n} = \frac{\prod_{i=1}^l (1+t_i)^n}{\prod_{i=1}^l (1+t_i^2)^m \prod_{1 \leq i < j \leq l} (1+t_i t_j)^{2m}}. \quad (4.11)$$

Note that the generating function $H_{m,n}$ can be regarded as the generalization of $T_{m,n}$ in the multi-homogeneous setting.

We state it in the following conjecture.

Conjecture 1. For a random (v, o, q, l, m) SNOVA scheme, with very high probability, we have

$$\dim_{\mathbb{F}_{q^l}}(\ker \bar{\mathcal{M}}_{\text{SNOVA}}|_{\bar{\mathcal{D}}}) = \max(H_{m,n}[\mathbf{t}^o], 1) \quad (4.12)$$

where $H_{m,n}[\mathbf{t}^o] = H_{m,n}[\mathbf{t}^{(o, \dots, o)}]$.

Note that $H_{m,n}[\mathbf{t}^o]$ is the coefficient of $\prod_{i=1}^l t_i^o$ in the series $H_{m,n}$. and

$$\dim_{\mathbb{F}_{q^l}}(\ker \bar{\mathcal{M}}_{\text{SNOVA}}|_{\bar{\mathcal{D}}}) = \dim_{\mathbb{F}_q}(\ker \mathcal{M}_{\text{SNOVA}}|_{\mathcal{D}}). \quad (4.13)$$

Similar to the case of Theorem 3, we can verify the correctness of $H_{m,n}$ by checking whether the coefficient $H_{m,n}[\mathbf{t}^o]$ matches the dimension of the kernel of the wedge map observed in the experimental data in Section 3.4. We present the prediction together with the experimental data in the following Table 2.

From Table 2, we observe that the generating function proposed above provides accurate estimates of the corresponding $\dim(\ker \bar{\mathcal{M}}_{\text{SNOVA}}|_{\bar{\mathcal{D}}})$ in our Magma experiments. Note that the number with “*” in Table 2 is different from the reported data by Le *et al.* [9]. We believe that the number they reported might be a typo. The remaining data are consistent with those reported in their paper.

Table 2. Table of the kernel dimensions $\dim(\ker \bar{\mathcal{M}}_{\text{SNOVA}}|_{\bar{\mathcal{D}}})$ obtained by our Magma implementation together with the corresponding values estimated from the generating function.

(v, o, q, l, m, n)	$\dim(\ker \bar{\mathcal{M}}_{\text{SNOVA}} _{\bar{\mathcal{D}}})$	$\max(H_{m,n}[\mathbf{t}^o], 1)$
(3, 2, 16, 2, 1, 5)	34	34
(4, 2, 16, 2, 1, 6)	127	127
(4, 2, 16, 2, 2, 6)	35	35
(4, 3, 16, 2, 1, 7)	127	127
(5, 2, 16, 2, 1, 7)	305	305
(2, 1, 16, 3, 1, 3)	9	9
(3, 1, 16, 3, 1, 4)	40	40
(2, 1, 16, 4, 1, 3)	1	1

Table 3. Table of the rank of the wedge map obtained by our Magma implementation. Note that the number with “*” in the table is different from the reported data by Le *et al.* [9]. We believe that the number they reported might be a typo.

(v, o, q, l, m, n)	$\dim_{\mathbb{F}_q}(\tilde{D}) = \binom{n}{v}^l$	$\text{rank}(\tilde{\mathcal{M}}_{\text{SNOVA}} _{\tilde{D}})$
(3, 2, 16, 2, 1, 5)	100	66
(4, 2, 16, 2, 1, 6)	225	98
(4, 2, 16, 2, 2, 6)	225	190
(4, 3, 16, 2, 1, 7)	1225	1098*
(5, 2, 16, 2, 1, 7)	441	136
(2, 1, 16, 3, 1, 3)	27	18
(3, 1, 16, 3, 1, 4)	64	24
(2, 1, 16, 4, 1, 3)	81	80

For consistency with the work of Le *et al.* [9], we also list the numbers of $\text{rank}(\tilde{\mathcal{M}}_{\text{SNOVA}}|_{\tilde{D}})$ in Table 3. It is worth noting that both estimation approaches (i.e., the formula (8) and (9)) presented in the work of Le *et al.* [9] fail to give a truly exact prediction of the kernel dimension $\dim(\ker \tilde{\mathcal{M}}_{\text{SNOVA}}|_{\tilde{D}})$.

4.2 Complexity Estimation

For the complexity estimation, we assume that random memory access is free and that the cost of finding a kernel vector of a $N \times N$ matrix over $\mathbb{F}_q$ with density ρ using sparse linear algebra has a cost of

$$3 \cdot \rho \cdot N^2 \cdot (2 \log_2(q)^2 + \log_2(q)). \quad (4.14)$$

The matrix on which we will perform the sparse linear algebra is the wedge map corresponding to $\text{SNOVA}(v, o', l, m, q)$ where

$$o' = \min \{ o' \mid 1 < o' \leq o, H_{m,n'}[\mathbf{t}^{o'}] \leq 1 \}. \quad (4.15)$$

and $n' = v + o'$. This matrix has size with the density of

$$N \times N = \binom{v+o'}{v}^l \times \binom{v+o'}{v}^l, \quad \rho = \binom{lv+2}{2}. \quad (4.16)$$

This results in a cost (in gates) of

$$3 \cdot \binom{lv+2}{2} \binom{v+o'}{v}^{2l} (2 \log_2(q)^2 + \log_2(q)). \quad (4.17)$$

The following table presents the complexity estimates for the SNOVA parameter sets. According to Table 4, we show that six parameter sets fall below NISTs required security levels. In the Table 5, we present several representative parameter sets by slightly increasing number of vinegar variables. As is consistent

with other UOV variants, a minor increase in the number of vinegar variables is sufficient to resist the wedge attack. This is precisely because our results allow us to accurately estimate the dimension of the kernel of the wedge map.

Consequently, we obtain complexity estimates that are more convincing than those presented in previous works [9]. We believe that this captures the spirit of cryptanalysis: through a security analysis of SNOVA, we gain a deeper understanding of the impact of the wedge attack, and at the same time, we are able to propose well-justified parameter adjustments. In Table 4, it is worth noting that the all $l = 4$ parameter sets continue to meet the targeted security requirements. The $l = 5$ parameter sets need further studies, we leave it as future work.

Table 4. Complexity estimates of SNOVA parameter sets against wedge attacks in $\log_2(\# \text{gates})$. The notation **N/A** denotes that the attack is not applicable.

SL	(v, o, q, l, m)	o'	complexity
I	(37, 17, 16, 2, 17)	8	128
	(25, 8, 16, 3, 8)	5	120
	(24, 5, 16, 4, 5)	5	153
III	(56, 25, 16, 2, 25)	11	180
	(49, 11, 16, 3, 11)	11	250
	(37, 8, 16, 4, 8)	7	221
	(24, 5, 16, 5, 5)	4	162
V	(75, 33, 16, 2, 33)	13	220
	(66, 15, 16, 3, 15)	14	323
	(60, 10, 16, 4, 10)	N/A	N/A
	(29, 6, 16, 5, 6)	5	201

Table 5. Demonstrative SNOVA parameter set adjustments.

SL	(v, o, q, l, m)	$(v_{\text{new}}, o, q, l, m)$
I	(37, 17, 16, 2, 17)	(48, 17, 16, 2, 17)
	(25, 8, 16, 3, 8)	(32, 8, 16, 3, 8)
III	(56, 25, 16, 2, 25)	(73, 25, 16, 2, 25)
V	(75, 33, 16, 2, 33)	(98, 33, 16, 2, 33)

5 Rank of $\mathcal{M}$ via Complex

In this section, we will present the calculations for $\dim \ker \mathcal{M}_{\text{UOV}}$ defined in [13], and $\dim \ker \mathcal{M}_{\text{SNOVA}}$ defined in this paper via the homology theory.

5.1 Character and Characteristic of Complex

In this subsection, we will introduce the definitions of character and characteristic of complex.

Definition 1 (Character of a graded module). Let $M = \bigoplus_{i \in \mathbb{Z}} M_i$ be a $\mathbb{Z}$ -graded vector space over a field, where each M_i is finite-dimensional. The character of M is the formal power series

$$\text{ch}(M) := \sum_{i \in \mathbb{Z}} (\dim M_i) t^i.$$

More generally, if M is $\mathbb{Z}^l$ -graded,

$$M = \bigoplus_{\mathbf{a} \in \mathbb{Z}^l} M_{\mathbf{a}}, \quad \mathbf{a} = (a_1, \dots, a_l),$$

then its character is

$$\text{ch}(M) := \sum_{\mathbf{a} \in \mathbb{Z}^l} (\dim M_{\mathbf{a}}) \mathbf{t}^{\mathbf{a}}, \quad \mathbf{t}^{\mathbf{a}} := t_1^{a_1} \cdots t_l^{a_l}.$$

For convention, we set

$$\text{ch}(M_i) = t^i \cdot \dim M_i \quad \text{and} \quad \text{ch}(M_{i_1, \dots, i_l}) = t_1^{i_1} \cdots t_l^{i_l} \cdot \dim M_{i_1, \dots, i_l}.$$

Definition 2 (Character of a complex). Let $C^\bullet$ be a cochain complex of graded vector spaces with finite dimensional graded components and of finite length

$$\cdots \rightarrow C^{i-1} \xrightarrow{d_{i-1}} C^i \xrightarrow{d_i} C^{i+1} \rightarrow \cdots.$$

Its character is defined as the alternating sum

$$\text{ch}(C^\bullet) := \sum_{i \in \mathbb{Z}} (-1)^i \text{ch}(C^i).$$

If each C^i is $\mathbb{Z}^l$ -graded, then

$$\text{ch}(C^\bullet) = \sum_{i \in \mathbb{Z}} (-1)^i \text{ch}(C^i) = \sum_{\mathbf{a} \in \mathbb{Z}^l} \left(\sum_{i \in \mathbb{Z}} (-1)^i \dim C_{\mathbf{a}}^i \right) \mathbf{t}^{\mathbf{a}}.$$

Definition 3 (characteristic). Let $C^\bullet$ be a cochain complex of finite-dimensional vector spaces and of finite length

$$\cdots \rightarrow C^{i-1} \xrightarrow{d_{i-1}} C^i \xrightarrow{d_i} C^{i+1} \rightarrow \cdots.$$

The characteristic of $C^\bullet$ is defined as

$$\chi(C^\bullet) := \sum_{i \in \mathbb{Z}} (-1)^i \dim C^i.$$

Example 1. Let $M = \bigoplus_{i=1}^l V_i$, where each $V_i \cong \mathbb{F}_q^n$, and assign multi-degree $\mathbf{e}_i \in \mathbb{Z}^l$ to all elements of V_i (i.e. degree 1 in the i -th component and 0 elsewhere).

Since the exterior algebra preserves direct sums, we have

$$\wedge M \cong \bigwedge \left(\bigoplus_{i=1}^l V_i \right) \cong \bigotimes_{i=1}^l \wedge V_i,$$

where the tensor product carries the induced $\mathbb{Z}^l$ -grading. Therefore, we have

$$\text{ch}(M) = \sum_{i_1, \dots, i_l \geq 0} \binom{n}{i_1} \cdots \binom{n}{i_l} t_1^{i_1} \cdots t_l^{i_l} = \prod_{i=1}^l (1 + t_i)^n.$$

Example 2. Let

$$V = \mathbb{F}_q^n \cong \bigoplus_{i=1}^n \mathbb{F}_q e_i, \quad W = \mathbb{F}_q^m \cong \bigoplus_{i=1}^m \mathbb{F}_q \eta_i,$$

with $\deg(e_i) = 1, \deg(\eta_i) = 2$ and $\text{char } \mathbb{F}_q = 2$. Fix m alternating forms $q_i \in \wedge^2 V$ for $i \in [m]$.

For $r \geq 0$ define

$$D^r := \wedge V \otimes S^r W,$$

where $S^r W$ is the r -th symmetric power of W , and the differential

$$d_r : D^r \rightarrow D^{r+1}, \quad d_r(\alpha \otimes \omega) = \sum_{i=1}^m (\alpha \wedge q_i) \otimes (\eta_i \cdot \omega),$$

for $\alpha \in \wedge V$ and $\omega \in S^r W$. Then $(D^\bullet, d_\bullet)$ is a complex. Indeed,

$$\begin{aligned} d_{r+1} d_r(\alpha \otimes \omega) &= d_{r+1} \left(\sum_{i=1}^m (\alpha \wedge q_i) \otimes (\eta_i \omega) \right) \\ &= \sum_{i=1}^m \sum_{j=1}^m (\alpha \wedge q_i \wedge q_j) \otimes (\eta_j \eta_i \omega) \\ &= 2 \sum_{1 \leq i < j \leq m} (\alpha \wedge q_i \wedge q_j) \otimes (\eta_j \eta_i \omega) = 0, \end{aligned}$$

where the last equality holds because $\text{char}(\mathbb{F}_q) = 2$. Therefore the character of $(D^\bullet, d_\bullet)$ is

$$\begin{aligned} \text{ch}(D^\bullet) &= \sum_{r \geq 0} (-1)^r \text{ch}(D^r) \\ &= \sum_{r \geq 0} (-1)^r \text{ch}(\wedge V) \cdot \text{ch}(S^r W) \\ &= (1+t)^n \cdot \sum_{r \geq 0} (-1)^r \binom{m+r-1}{r} t^{2r} \\ &= \frac{(1+t)^n}{(1+t^2)^m}. \end{aligned}$$

Example 3. Let

$$V = \bigoplus_{i=1}^l V_i, \quad V_i \cong \mathbb{F}_q^n, \quad \text{char } \mathbb{F}_q = 2,$$

and assign to each element of V_i the multi-degree $\mathbf{e}_i$ (with a 1 in the i -th position and 0 elsewhere). For each $i \in [l]$, let

$$W_i \cong \mathbb{F}_q^m$$

have multi-degree $2\mathbf{e}_i$, and for each $1 \leq i < j \leq l$, let

$$W_{ij} \cong \mathbb{F}_q^{2m}$$

have multi-degree $\mathbf{e}_i + \mathbf{e}_j$. Define

$$W = \bigoplus_{i=1}^l W_i \oplus \bigoplus_{1 \leq i < j \leq l} W_{ij}.$$

Fix ml^2 alternating forms

$$q_{k,i,j} \in \wedge^2 V$$

of multi-degree $\mathbf{e}_i + \mathbf{e}_j$, for all $k \in [m]$ and $1 \leq i, j \leq l$.

Let $\{e_{k,i,i} : k \in [m]\}$ be a basis of W_i , and let $\{e_{k,i,j} : k \in [2m]\}$ be a basis of W_{ij} for $1 \leq i < j \leq l$. Then the set

$$\{e_{k,i,i} : k \in [m], i \in [l]\} \cup \{e_{k,i,j} : k \in [2m], 1 \leq i < j \leq l\}$$

forms a basis of W .

Define

$$C^r := \wedge V \otimes S^r W$$

for $r \geq 0$, and the differential

$$d_r : C^r \longrightarrow C^{r+1}, \quad d_r(\alpha \otimes \omega) = \sum_{k,i,j} (\alpha \wedge q_{k,i,j}) \otimes (e_{k,i,j} \cdot \omega)$$

for $\alpha \in \wedge V$ and $\omega \in S^r W$. As the same as in Example 2, the sequence $(C^\bullet, d_\bullet)$ is a complex. Its character is

$$\begin{aligned} \text{ch}(C^\bullet) &= \sum_{r \geq 0} (-1)^r \text{ch}(C^r) \\ &= \sum_{r \geq 0} (-1)^r \text{ch}(\wedge V) \cdot \text{ch}(S^r W) \\ &= \prod_{i=1}^l (1 + t_i)^n \cdot \sum_{r \geq 0} (-1)^r \text{ch}(S^r W). \end{aligned} \tag{5.1}$$

To compute the second factor, observe that

$$\begin{aligned}
\sum_{r \geq 0} \text{ch}(S^r W) u^r &= \prod_{e \in \text{basis}(W)} \left(1 + \mathbf{t}^{\deg(e)} u + \mathbf{t}^{2 \deg(e)} u^2 + \dots \right) \\
&= \prod_{e \in \text{basis}(W)} \frac{1}{1 - u \mathbf{t}^{\deg(e)}} \\
&= \prod_{e \in W_i} \frac{1}{1 - ut_i^2} \cdot \prod_{e \in W_{ij}} \frac{1}{1 - ut_i t_j} \\
&= \prod_{i=1}^l \frac{1}{(1 - ut_i^2)^m} \cdot \prod_{1 \leq i < j \leq l} \frac{1}{(1 - ut_i t_j)^{2m}},
\end{aligned} \tag{5.2}$$

where the product ranges over the basis $\{e_{k,i,j}\}$ of W . Substituting $u = -1$ in Eq. (5.2) and combining with Eq. (5.1), we obtain

$$\begin{aligned}
\text{ch}(C^\bullet) &= \prod_{i=1}^l (1 + t_i)^n \cdot \sum_{r \geq 0} (-1)^r \text{ch}(S^r W) \\
&= \frac{\prod_{i=1}^l (1 + t_i)^n}{\prod_{i=1}^l (1 + t_i^2)^m \cdot \prod_{1 \leq i < j \leq l} (1 + t_i t_j)^{2m}}.
\end{aligned}$$

5.2 The Complex in Wedge Attack

In this subsection, we extend the maps $\mathcal{M}_{\text{UOV}}$ and $\bar{\mathcal{M}}_{\text{SNOVA}}$ to complexes, and compute the dimensions of their kernels via the cohomology of these complexes.

Complex in UOV Let $p_1, \dots, p_m \in \mathbb{F}_q[x_1, \dots, x_n]$ be the m public quadratic forms appearing in the UOV scheme, and let $q_1, \dots, q_m \in \wedge^2 V$ be the 2-form corresponding to the polar forms of $p_1 \dots, p_m$, where $V = \mathbb{F}_q^n$.

Let o be the dimension of the oil space in UOV and $v = n - o$. Consider the linear map

$$\mathcal{M}_{\text{UOV}} : \wedge^v V \longrightarrow (\wedge^{v+2} V)^m, \quad \omega \longmapsto (q_1 \wedge \omega, \dots, q_m \wedge \omega).$$

This map is the first differential of a truncated complex

$$(D_v^\bullet, d_\bullet),$$

defined by

$$D_v^r := \wedge^{v+2r} V \otimes S^r W, \quad r \geq 0,$$

where $W \cong \mathbb{F}_q^m$ with basis $\{\eta_1, \dots, \eta_m\}$. The differential

$$d_r : D_v^r \longrightarrow D_v^{r+1}$$

is induced from the full complex $(C^\bullet, d_\bullet)$ of Example 2:

$$d_r(\alpha \otimes \omega) = \sum_{i=1}^m (\alpha \wedge q_i) \otimes (\eta_i \cdot \omega), \quad \alpha \in \wedge^{v+2r} V, \omega \in S^r W.$$

One can check that d_0 coincides with $\mathcal{M}_{\text{UOV}}$ after identifying $(\wedge^{v+2} V)^m$ with C_v^1 and $D_v^\bullet \subset C^\bullet$ is a subcomplex.

Recall that the character of the full complex is the multivariate series

$$\text{ch}(D^\bullet) := \sum_{r \geq 0} (-1)^r \text{ch}(D^r) \quad \text{with} \quad \text{ch}(D^r) = \text{ch}(\wedge^r V) \cdot \text{ch}(S^r W),$$

as in Example 2. Taking the coefficient of t^o yields

$$\begin{aligned} \text{ch}(D^\bullet)[t^o] &= \sum_{r \geq 0} (-1)^r \dim(\wedge^{o-2r} V) \dim(S^r W) \\ &= \sum_{r \geq 0} (-1)^r \dim(\wedge^{v+2r} V) \dim(S^r W) \\ &= \chi(D_v^\bullet). \end{aligned} \tag{5.3}$$

Complex in SNOVA In the following, we will construct the $\tilde{\mathbf{M}}_{\text{SNOVA}}$ explicitly. Suppose that $p_{i,a,b}(\mathbf{x}) = \mathbf{x}^\top (\mathbf{S}^{\otimes n})^a \cdot \mathbf{P}_i \cdot (\mathbf{S}^{\otimes n})^b \mathbf{x}$. For any $\mathbf{x} \in \mathbb{F}_{q^l}^{nl}$, we can write

$$\mathbf{x} = \sum_{i=1}^l \mathbf{x}_i \otimes \tau^{i-1}(\xi), \quad \mathbf{x}_i \in \mathbb{F}_{q^l}^n. \tag{5.4}$$

Then we have

$$\begin{aligned} p_{i,a,b}(\mathbf{x}) &= \left(\sum_{r=1}^l \mathbf{x}_r \otimes \tau^{r-1}(\xi) \right)^\top \cdot (\mathbf{S}^{\otimes n})^a \cdot \mathbf{P}_i \cdot (\mathbf{S}^{\otimes n})^b \cdot \left(\sum_{s=1}^l \mathbf{x}_s \otimes \tau^{s-1}(\xi) \right) \\ &= \sum_{r=1}^l \sum_{s=1}^l \tau^{s-1}(\lambda^a) \tau^{r-1}(\lambda^b) \cdot \left(\mathbf{x}_s \otimes \tau^{s-1}(\xi) \right)^\top \cdot \mathbf{P}_i \cdot \left(\mathbf{x}_r \otimes \tau^{r-1}(\xi) \right) \end{aligned} \tag{5.5}$$

Define

$$\bar{p}_{i,r,s}(\mathbf{x}_1, \dots, \mathbf{x}_l) := \left(\mathbf{x}_s \otimes \tau^{s-1}(\xi) \right)^\top \cdot \mathbf{P}_i \cdot \left(\mathbf{x}_r \otimes \tau^{r-1}(\xi) \right). \tag{5.6}$$

Let $\bar{q}_{k,i,j} \in \wedge^2 V$ be the 2-forms corresponding to the polar forms defined by $\bar{p}_{k,i,j}$. In the following, we will construct the mapping of $\tilde{\mathcal{M}}_{\text{SNOVA}}$ explicitly.

Let

$$V = \bigoplus_{i=1}^l V_i, \quad V_i \cong \mathbb{F}_{q^l}^n,$$

and assign $\deg(V_i) = \mathbf{e}_i \in \mathbb{Z}_{\geq 0}^l$. After a suitable change of variables, one verifies that

$$\deg(\bar{q}_{k,i,j}) = \mathbf{e}_i + \mathbf{e}_j \quad (k \in [m], i, j \in [l]).$$

Let W be the multigraded vector space from Example 3 with basis $\{e_{k,i,j}\}$ and set

$$\deg(e_{k,i,j}) = \deg(q_{k,i,j}) = \mathbf{e}_i + \mathbf{e}_j.$$

Write

$$\mathbf{v} := (v, \dots, v) \in \mathbb{Z}_{\geq 0}^l, \quad \mathbf{o} := (o, \dots, o) \in \mathbb{Z}_{\geq 0}^l, \quad \text{so that } \mathbf{v} + \mathbf{o} = (n, \dots, n).$$

Consider the linear map

$$\bar{\mathcal{M}}_{\text{SNOVA}} : \wedge^{\mathbf{v}} V \longrightarrow (\wedge^{\mathbf{v} + \mathbf{e}_i + \mathbf{e}_j} V)_{k,i,j}, \quad \omega \longmapsto (\bar{q}_{k,i,j} \wedge \omega)_{k,i,j},$$

where we view $\wedge^{\mathbf{v}} V := \wedge^v V_1 \otimes \dots \otimes \wedge^v V_l$ as a subspace of $\wedge^{lv} V$. Indeed, one can check that $\wedge^{\mathbf{v}} V \cong \bar{\mathcal{D}}$ defined in Eq. (3.27).

Define the truncated complex $(C_{\mathbf{v}}^{\bullet}, d_{\bullet})$ by

$$C_{\mathbf{v}}^r := \bigoplus_{\mathbf{b} \in \mathbb{Z}_{\geq 0}^l} (\wedge^{\mathbf{v} + \mathbf{b}} V) \otimes (S^r W)_{\mathbf{b}}, \quad r \geq 0,$$

where $(S^r W)_{\mathbf{b}}$ denotes the $\mathbf{b}$ -homogeneous component of $S^r W$. The differential

$$d_r : C_{\mathbf{v}}^r \longrightarrow C_{\mathbf{v}}^{r+1}$$

is induced from the full complex $(C^{\bullet}, d_{\bullet})$ of Example 3:

$$d_r(\alpha \otimes \omega) = \sum_{k,i,j} (\alpha \wedge \bar{q}_{k,i,j}) \otimes (e_{k,i,j} \cdot \omega), \quad \alpha \in \wedge V, \omega \in S^r W.$$

In particular, since $S^0 W \cong \mathbb{F}_{q^t}$ and $(S^0 W)_{\mathbf{0}} = S^0 W$, we have

$$C_{\mathbf{v}}^0 = \wedge^{\mathbf{v}} V \otimes (S^0 W)_{\mathbf{0}} \cong \wedge^{\mathbf{v}} V,$$

and one checks that d_0 coincides with $\bar{\mathcal{M}}_{\text{SNOVA}}$ after identifying the codomain with coordinates in the basis $\{e_{k,i,j}\}$:

$$d_0(\omega) = \sum_{k,i,j} (\bar{q}_{k,i,j} \wedge \omega) \otimes e_{k,i,j}.$$

Hence $\ker(\bar{\mathcal{M}}_{\text{SNOVA}}) = \ker(d_0)$. One can verify that $C_{\mathbf{v}}^{\bullet} \subset C^{\bullet}$ is a (multigraded) subcomplex.

Recall that the character of the full complex is the multivariate series

$$\text{ch}(C^{\bullet}) := \sum_{r \geq 0} (-1)^r \text{ch}(C^r) \quad \text{with} \quad \text{ch}(C^r) = \text{ch}(\wedge V) \cdot \text{ch}(S^r W),$$

as in Example 3. Taking the coefficient of $\mathbf{t}^{\mathbf{o}}$ yields

$$\text{ch}(C^\bullet)[\mathbf{t}^{\mathbf{o}}] = \sum_{r \geq 0} (-1)^r \sum_{\mathbf{a}, \mathbf{b} \in \mathbb{Z}_{\geq 0}^l, \mathbf{a} + \mathbf{b} = \mathbf{o}} \dim(\wedge^{\mathbf{a}} V) \cdot \dim((S^r W)_{\mathbf{b}}). \quad (5.7)$$

Now use the blockwise binomial symmetry on each V_i :

$$\dim(\wedge^{a_i} V_i) = \binom{n}{a_i} = \binom{n}{n - a_i} = \dim(\wedge^{n - a_i} V_i),$$

which implies for $\mathbf{a} = (a_1, \dots, a_l)$ that

$$\dim(\wedge^{\mathbf{a}} V) = \prod_{i=1}^l \dim(\wedge^{a_i} V_i) = \prod_{i=1}^l \dim(\wedge^{n - a_i} V_i) = \dim(\wedge^{(n, \dots, n) - \mathbf{a}} V)$$

Taking $\mathbf{a} = \mathbf{o} - \mathbf{b}$ and using $\mathbf{v} + \mathbf{o} = (n, \dots, n)$, we get

$$\dim(\wedge^{\mathbf{o} - \mathbf{b}} V) = \dim(\wedge^{\mathbf{v} + \mathbf{b}} V).$$

Substituting into Eq. (5.7) gives the conceptual identification

$$\begin{aligned} \text{ch}(C^\bullet)[\mathbf{t}^{\mathbf{o}}] &= \sum_{r \geq 0} (-1)^r \sum_{\mathbf{b} \in \mathbb{Z}_{\geq 0}^l} \dim(\wedge^{\mathbf{v} + \mathbf{b}} V) \dim((S^r W)_{\mathbf{b}}) \\ &= \sum_{r \geq 0} (-1)^r \dim(C_{\mathbf{v}}^r) \\ &= \chi(C_{\mathbf{v}}^\bullet). \end{aligned} \quad (5.8)$$

Lemma 3. *Suppose we have a complex of finite dimensional vector spaces over a field:*

$$0 \xrightarrow{d_{-1}} C_0 \xrightarrow{d_0} C_1 \xrightarrow{d_1} \dots \xrightarrow{d_{m-1}} C_m \xrightarrow{d_m} 0, \quad (5.9)$$

where the linear map $\{d_i : i = -1, \dots, m\}$ satisfy $d_{i+1} \circ d_i = 0$, d_{-1} and d_m are zero maps. If the complex is exact, i.e. $\text{im} d_{i-1} = \ker d_i$ for each $i \in [m]$, then we have

$$\sum_{r=0}^m (-1)^r \dim C_r = 0.$$

Proof. Indeed, when $m = 2$, the statement holds trivially. Then the general case follows from the following short exact sequences

$$0 \rightarrow \ker d_r \rightarrow C_r \rightarrow \text{im} d_r \rightarrow 0, \quad \text{for } r = 0, 1, \dots, m-1,$$

Note that the exactness implies that $\text{im} d_r = \ker d_{r+1}$. □

By Lemma 3 above, we have

Theorem 4. *For UOV, if the complex $D_v^\bullet$ is exact everywhere except at the zeroth position, then*

$$\begin{aligned} \dim \ker \mathcal{M}_{\text{UOV}} &= \dim H^0(D_v^\bullet) = \sum_{r \geq 0} (-1)^r \dim(D_v^r) \\ &= \frac{(1+t)^n}{(1+t^2)^m} [t^o]. \end{aligned}$$

For SNOVA, if the complex $C_v^\bullet$ is exact everywhere except at the zeroth position, then

$$\begin{aligned} \dim \ker \bar{\mathcal{M}}_{\text{SNOVA}} &= \dim H^0(C_v^\bullet) = \sum_{r \geq 0} (-1)^r \dim(C_v^r) \\ &= \frac{\prod_{i=1}^l (1+t_i)^n}{\prod_{i=1}^l (1+t_i^2)^m \cdot \prod_{1 \leq i < j \leq l} (1+t_i t_j)^{2m}} [\mathbf{t}^{(o, \dots, o)}]. \end{aligned}$$

Remark 1. We conjecture that for random UOV and SNOVA system, the complex $D_v^\bullet$ and $C_v^\bullet$ satisfy the condition in the above theorem.

5.3 Reduction to Semi-Regular Assumption

In this subsection, we follow the proof of [13, Proposition], give a similar explanation of our formula.

Assumption on semi-regular in multi-homogeneous system. Let $\text{char } \mathbb{F}_q = 2$,

$$f_1, \dots, f_m \in \mathbb{F}_q[\mathbf{x}_1, \dots, \mathbf{x}_l], \quad \mathbf{x}_i = (x_{i1}, \dots, x_{in}),$$

be multi-homogeneous polynomials with respect to the variable sets $\mathbf{x}_1, \dots, \mathbf{x}_l$. Suppose that $\deg(f_i) = \mathbf{d}_i \in \mathbb{Z}_{\geq 0}^l$ for $i \in [m]$.

Then we assume that the Hilbert series of the quotient ring

$$\mathbb{F}_q[\mathbf{x}_1, \dots, \mathbf{x}_l] / \langle x_{ij}^2, f_1, \dots, f_m \rangle$$

is given by

$$\frac{\prod_{i=1}^l (1+t_i)^n}{\prod_{i=1}^m (1+\mathbf{t}^{\mathbf{d}_i})}.$$

when $f_1, \dots, f_m$ is a semi-regular sequence. This assumption is a natural extension of the semi-regularity assumption in the homogeneous case.

Proposition 1. *Under the above assumption. Let $\bar{p}_{k,i,j}$ be defined as above. If the polynomials $p_1, \dots, p_m$ form a semi-regular sequence in $\mathbb{F}_{q^l}[\mathbf{x}_1, \dots, \mathbf{x}_n] / \langle x_{ij}^2 \rangle$, then the linear map $\bar{\mathcal{M}}_{\text{SNOVA}}$ has a kernel of dimension*

$$\frac{\prod_{i=1}^l (1+t_i)^n}{\prod_{i=1}^l (1+t_i^2)^m \cdot \prod_{1 \leq i < j \leq l} (1+t_i t_j)^{2m}} [\mathbf{t}^{(o, \dots, o)}]. \quad (5.10)$$

Proof. First, we define an isomorphism of the vector spaces:

$$\bigwedge^k V \cong \bigwedge^{n-k} V, \quad \mathbf{e}_I \mapsto \mathbf{e}_{[n] \setminus I},$$

where $\mathbf{e}_I = \bigwedge_{i \in I} e_i$ and $e_1, \dots, e_n$ is a basis of V . Note that $e_i \wedge e_j = e_j \wedge e_i$ when $\text{char}(\mathbb{F}_q) = 2$.

Let us construct the map

$$\tilde{\mathcal{M}}_{\text{SNOVA}} : (\wedge^{\mathbf{o} - \mathbf{e}_i - \mathbf{e}_j} V)_{k,i,j} \longrightarrow \wedge^{\mathbf{o}} V \quad (\omega_{k,i,j}) \mapsto \sum_{k,i,j} \bar{q}_{k,i,j} \wedge \omega_{k,i,j}. \quad (5.11)$$

We now combine this map with the isomorphism to obtain

$$\bar{\mathcal{M}}_{\text{SNOVA}}^{\vee} : (\wedge^{\mathbf{v} + \mathbf{e}_i + \mathbf{e}_j} V)_{k,i,j} \rightarrow (\wedge^{\mathbf{o} - \mathbf{e}_i - \mathbf{e}_j} V)_{k,i,j} \xrightarrow{\tilde{\mathcal{M}}_{\text{SNOVA}}} \wedge^{\mathbf{o}} V \rightarrow \wedge^{\mathbf{o}} V. \quad (5.12)$$

In the same way as in [13, Proposition 4.2], one can show that $\bar{\mathcal{M}}_{\text{SNOVA}}^{\vee} = \bar{\mathcal{M}}_{\text{SNOVA}}^T$, and hence $\text{rank}(\bar{\mathcal{M}}_{\text{SNOVA}}) = \text{rank}(\bar{\mathcal{M}}_{\text{SNOVA}}^{\vee})$.

Next, we construct an isomorphism of algebras

$$\bigwedge_{i=1}^l \bigoplus V_i \rightarrow \mathbb{F}_{q^l}[\mathbf{x}_1, \dots, \mathbf{x}_l] / \langle x_{ij}^2 \rangle, \quad \mathbf{e}_{I_1} \otimes \dots \otimes \mathbf{e}_{I_l} \mapsto \prod_{i=1}^l \prod_{j \in I_i} x_{ij}. \quad (5.13)$$

Let $\tilde{\bar{p}}_{k,i,j}$ be the image of $\bar{q}_{k,i,j}$ under this isomorphism. Then one can check that $\tilde{\bar{p}}_{k,i,j}$ is exactly the quadratic form $\bar{p}_{k,i,j}$ considered module the ideal $\langle x_{ij}^2 \rangle$.

Furthermore, under this isomorphism, the map $\tilde{\mathcal{M}}_{\text{SNOVA}}$ is exactly the Macaulay matrix of the polynomials $\bar{p}_{k,i,j}$ of degree $\mathbf{o}$ in $\mathbb{F}_{q^l}[\mathbf{x}_1, \dots, \mathbf{x}_n] / \langle x_{ij}^2 \rangle$. Hence, if $(\bar{p}_{k,i,j})$ is semi-regular, under the above assumption, we can determine the rank of $\tilde{\mathcal{M}}_{\text{SNOVA}}$ by the Hilbert series of the quotient ring $\mathbb{F}_{q^l}[\mathbf{x}_1, \dots, \mathbf{x}_n] / \langle x_{ij}^2, \bar{p}_{k,i,j} \rangle$, which is

$$\frac{\prod_{i=1}^l (1 + t_i)^n}{\prod_{i=1}^l (1 + t_i^2)^m \cdot \prod_{1 \leq i < j \leq l} (1 + t_i t_j)^{2m}}. \quad (5.14)$$

This completes the proof. $\square$

6 Conclusion

In this work, we have conducted a comprehensive investigation into the algebraic structure of SNOVA, establishing a rigorous theoretical framework for understanding its resistance to wedge map attacks. By employing lifting techniques, we demonstrated a natural isomorphic relation between the SNOVA scheme over the matrix ring and its representation over an extension field, proving that the rank of the wedge map is a structural invariant unaffected by base change. Furthermore, we derived a generating function that accurately predicts the kernel dimension, resolving the inaccuracies of previous heuristic formulas. This precise

algebraic analysis not only deepens our understanding of the underlying complex but also provides a deterministic method for evaluating the security of SNOVA parameters.

Our findings offer a significant advantage for the ongoing standardization process, placing the security of SNOVA on a solid mathematical foundation. We believe this will make the security analysis of SNOVA more complete. Building on these structural insights, we will introduce a refined SNOVA design that ensures security against all known attacks while achieving an even smaller public key size in the forthcoming work.

References

1. Ward Beullens. Improved cryptanalysis of SNOVA. In *Annual International Conference on the Theory and Applications of Cryptographic Techniques*, pages 277–293. Springer, 2025.
2. Ward Beullens, Fabio Campos, Sofia Celi, Basil Hess, and Matthias J. Kannwischer. Mayo. *NIST Computer Security Resource Center (CSRC)*, 2025.
3. Ward Beullens, Ming-Shing Chen, Jintai Ding, Boru Gong, Matthias J. Kannwischer, Jacques Patarin, Bo-Yuan Peng, Dieter Schmidt, Cheng-Jhih Shih, Chengdong Tao, and Bo-Yin Yang. Uov: Unbalanced oil and vinegar. *NIST Computer Security Resource Center (CSRC)*, 2025.
4. Daniel Cabarcas, Peigen Li, Javier Verbel, and Ricardo Villanueva-Polanco. Improved attacks for snova by exploiting stability under a group action. In Yael Tauman Kalai and Seny F. Kamara, editors, *Advances in Cryptology – CRYPTO 2025*, pages 358–389, Cham, 2025. Springer Nature Switzerland.
5. David S. Dummit and Richard M. Foote. *Abstract Algebra*. John Wiley & Sons, Hoboken, NJ, 3 edition, 2004.
6. Hiroki Furue, Yasuhiko Ikematsu, Fumitaka Hoshino, Tsuyoshi Takagi, Haruhisa Kosuge, Kimihiro Yamakoshi, Rika Akiyama, Satoshi Nakamura, Shingo Orihara, and Koha Kinjo. Qr-uov. *NIST Computer Security Resource Center (CSRC)*, 2025.
7. Hiroki Furue, Yasuhiko Ikematsu, Shuhei Nakamura, and Rika Akiyama. Improved cryptanalysis of snova by solving multi-homogeneous systems via matrix transformations. In Goichiro Hanaoka and Bo-Yin Yang, editors, *Advances in Cryptology – ASIACRYPT 2025*, pages 405–435, Singapore, 2026. Springer Nature Singapore.
8. Yasuhiko Ikematsu and Rika Akiyama. Revisiting the security analysis of SNOVA. In *Proceedings of the 11th ACM Asia Public-Key Cryptography Workshop*, pages 54–61, 2024.
9. Thai Hung Le, Maxime Bros, Jacob Lichtinger, Brice Minaud, Ray Perlner, Daniel Smith-Tone, and Cristian Valenzuela. Exploiting snova’s structure in the wedge product attack. *NIST Computer Security Resource Center (CSRC), Sixth PQC Standardization Conference*, 2025.
10. Peigen Li and Jintai Ding. Cryptanalysis of the SNOVA signature scheme. In *International Conference on Post-Quantum Cryptography*, pages 79–91. Springer, 2024.
11. Peigen Li, Hao Guo, and Jintai Ding. Unified approach to UOV-like multivariate signature schemes. *Cryptology ePrint Archive*, Paper 2025/1778, 2025.

12. Shuhei Nakamura, Yusuke Tani, and Hiroki Furue. Lifting approach against the snova scheme. In *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences, 2025, Volume E108.A, Issue 10*, pages 1373–1381, 2025.
13. Lars Ran. Wedges, oil, and vinegar – an analysis of UOV in characteristic 2. Cryptology ePrint Archive, Paper 2025/1143, 2025.
14. Toshihiro Suzuki, Hiroki Furue, Takuma Ito, Shuhei Nakamura, and Shigenori Uchiyama. An extended rectangular minrank attack against UOV and its variants. *Cryptology ePrint Archive*, 2025.
15. Lih-Chung Wang, Chun-Yen Chou, Jintai Ding, Yen-Liang Kuan, Jan Adriaan Leegwater, Ming-Siou Li, Bo-Shu Tseng, Po-En Tseng, and Chia-Chun Wang. A note on the snova security. *NIST Computer Security Resource Center (CSRC)*, 2025.
16. Lih-Chung Wang, Chun-Yen Chou, Jintai Ding, Yen-Liang Kuan, Jan Adriaan Leegwater, Ming-Siou Li, Bo-Shu Tseng, Po-En Tseng, and Chia-Chun Wang. On the security of round 2 snova. *NIST Computer Security Resource Center (CSRC)*, 2025.
17. Lih-Chung Wang, Chun-Yen Chou, Jintai Ding, Yen-Liang Kuan, Jan Adriaan Leegwater, Ming-Siou Li, Bo-Shu Tseng, Po-En Tseng, and Chia-Chun Wang. Snova. *NIST Computer Security Resource Center (CSRC)*, 2025.
18. Lih-Chung Wang, Po-En Tseng, Yen-Liang Kuan, and Chun-Yen Chou. A Simple Noncommutative UOV Scheme. *Taiwanese Journal of Mathematics*, pages 1–35, 2026.